

Hellenic Complex Systems Laboratory

Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

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Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

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Abstract

Background: Diagnostic tests are evaluated by their ability to discriminate between diseased and nondiseased individuals; however, estimates of diagnostic accuracy alone do not quantify their associated uncertainty. Uncertainty assessment is therefore essential for interpreting accuracy estimates and comparing diagnostic thresholds.

Methods: We develop a parametric framework for propagating uncertainty to diagnostic accuracy measures formulated as functions of sensitivity, specificity, and prevalence. Normal, lognormal, or gamma models are specified separately for the diseased and nondiseased populations. At a fixed threshold, first-order uncertainty propagation retains covariance between estimated population means and standard deviations, incorporates prevalence sampling uncertainty when applicable, and represents common measurement uncertainty through an equivalent threshold shift relative to a fixed diagnostic threshold. The Wolfram Language implementation provides uncertainty components by source, joint output covariance, and approximate expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions. At optimized thresholds, uncertainty is propagated through the nonlinear measurement model by Gauss–Hermite quadrature, conditional on the estimated optimal threshold.

Results: Illustrative analyses indicate that the contribution of each uncertainty source varies with the diagnostic accuracy measure, distributional assumptions, and threshold. Dependence induced by the common measurement bias can partially cancel its contribution to the uncertainty of a derived measure. In the cases examined, joint propagation reproduces the corresponding marginal variances and enables evaluation of uncertainty across diagnostic thresholds.

Conclusions: The framework provides a reproducible basis for assessing diagnostic accuracy and uncertainty while preserving relevant input and output dependencies. Decomposition by source helps identify the assumptions and uncertainty components underlying each estimate. The intervals remain local approximations; nominal coverage and additional uncertainty from threshold selection require separate assessment.

Keywords:

diagnostic accuracy; covariance propagation; measurement uncertainty; sampling uncertainty; diagnostic threshold; delta method; lognormal distribution; effective degrees of freedom; Wolfram Language

1. Introduction

Medical diagnosis integrates clinical evidence with a model of disease status [1]. Quantitative tests classify measurements relative to a diagnostic threshold. Their performance depends on the measurement distributions in diseased and nondiseased populations and on the population in which the test is applied [2–4].

Diseased and nondiseased populations are denoted by D and H. At zero nominal measurement bias, results are positive when $X \geq t$ and negative when $X < t$. Figure 1 illustrates overlapping population distributions; Table 1 defines the corresponding contingency table.

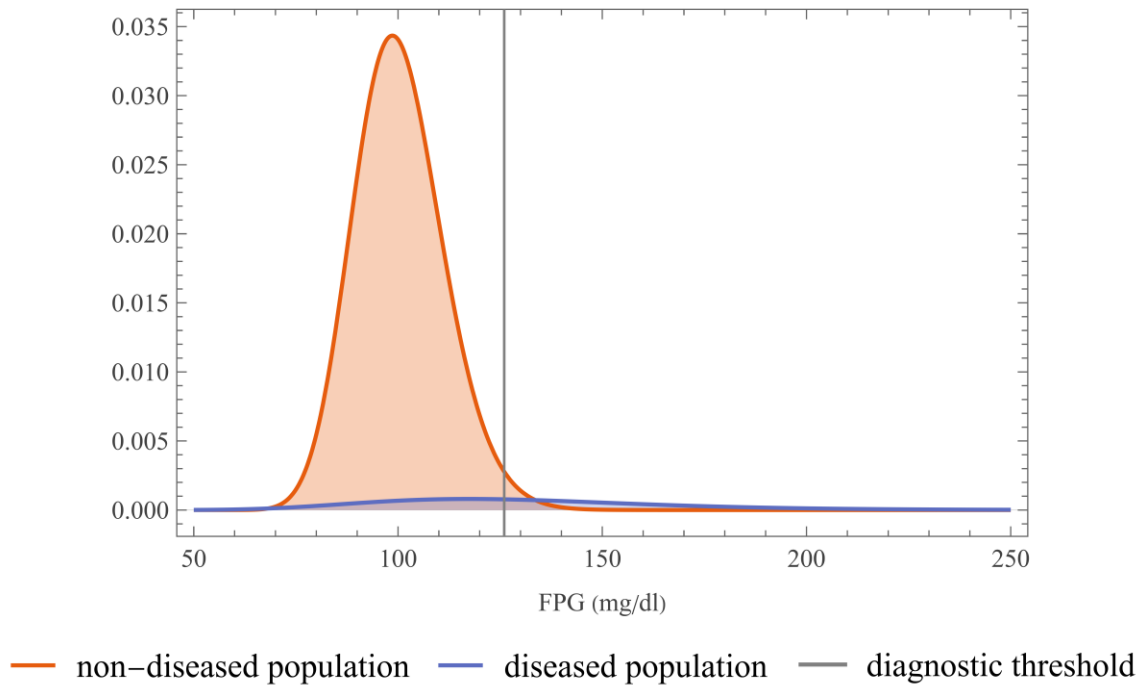


Figure 1. Lognormal probability densities for the supplied FPG inputs at an illustrative threshold of 126 mg/dL.

Table 1. Contingency table for a positive-above-threshold test.

Test result	Diseased (D)	Nondiseased (H)
Positive ($X \geq t$)	TP	FP
Negative ($X < t$)	FN	TN

1.1. Diagnostic accuracy measures

The sixteen measures are sensitivity (Se), specificity (Sp), overall diagnostic accuracy (ACC), positive and negative predictive values (PPV, NPV), diagnostic odds ratio (DOR), positive and negative likelihood ratios (LR^+ , LR^-), Youden's index (J), balanced accuracy (BA), concordance probability (CZ), Fowlkes–Mallows index (FMI), Cohen's kappa (κ), prevalence- and bias-adjusted kappa (PABAK), F_1 score (F_1), and Matthews correlation coefficient (MCC) [4, 5].

Table 2. Interpretation and admissible range of the sixteen diagnostic accuracy measures.

Measure	Interpretation	Range
Se	Proportion positive among diseased individuals.	[0, 1]
Sp	Proportion negative among nondiseased individuals.	[0, 1]
ACC	Prevalence-weighted probability of correct classification.	[0, 1]
PPV	Probability of disease conditional on a positive result.	[0, 1]
NPV	Probability of no disease conditional on a negative result.	[0, 1]
DOR	Ratio of the odds of a positive result in the two populations.	[0, ∞)
LR^+	Se divided by the false-positive probability.	[0, ∞)
LR^-	False-negative probability divided by Sp.	[0, ∞)
J	Se plus Sp minus one.	[-1, 1]
BA	Arithmetic mean of Se and Sp.	[0, 1]
CZ	Product of Se and Sp.	[0, 1]
FMI	Geometric mean of Se and PPV.	[0, 1]
κ	Agreement corrected for the chance agreement from both marginals.	[-1, 1]
PABAK	2 ACC – 1; generally depends on prevalence.	[-1, 1]
F_1	Harmonic mean of Se and PPV.	[0, 1]
MCC	Correlation between binary test classification and disease status.	[-1, 1]

Increasing the threshold decreases Se and increases Sp under this decision rule. This trade-off between detection and false positives does not alone determine a clinically optimal threshold. PPV and NPV also depend on prevalence, and operational decisions may depend on consequences and costs [3].

1.2. Uncertainty

Standard uncertainty expresses uncertainty on a standard-deviation scale. For a derived measure, specified input uncertainties and covariances are propagated through the measurement model [6, 7]. DAM uncertainty is therefore conditional on the model, sampling design, and represented uncertainty sources.

Parameter and prevalence sampling uncertainties constitute total sampling uncertainty. Combined standard uncertainty additionally includes common measurement uncertainty.

The measurement-uncertainty model represents an uncertain shift in measured values common to both populations. Its contribution is evaluated from both distribution tails at the fixed diagnostic

threshold while holding the fitted distributions fixed; parameter and prevalence sampling uncertainties are evaluated separately. A positive measurement bias increases Se and decreases Sp, with the reverse occurring for a negative bias. This common source induces negative covariance between the two rates, which must be retained when propagating uncertainty to measures that depend on both [6].

The diagnostic threshold t is a specified fixed decision value and has no assigned measurement uncertainty. Let Δ denote a measurement bias common to measured values in both populations. If X is the value before this shift and Y the resulting measured value, then $Y = X + \Delta$, classification by $Y \geq t$ is equivalent to $X \geq \tau$, where $\tau = t - \Delta$. The transformed threshold on the reference measurement scale τ is uncertain because of uncertainty in the measurement bias; its nominal value is t when $\Delta = 0$. The standard uncertainty $u_{\text{meas}}(t)$ quantifies uncertainty in this common measurement bias at the chosen cutoff. It does not describe uncertainty in the cutoff itself.

Independent perturbations of Se and Sp assign separate uncertainty sources and omit their dependence. Such a model requires separate justification, uncertainty magnitudes, and dependence assumptions. Even with identical marginal uncertainties, it generally yields different uncertainties for measures combining the two rates. The present model instead propagates their joint response to a common measurement bias, conditional on the specified population distributions.

Random analytical variation in individual measurements can alter the observed population distributions and inferred diagnostic performance. Under an independent additive model, each observed distribution is the convolution of the underlying population distribution with the distribution of analytical variation. This may affect distribution tails, the ROC curve, and the estimated optimal threshold. Analysis requires an explicit model specifying the mechanism, magnitude, and dependence structure of analytical variation. It must also distinguish underlying distributions that exclude analytical variation from observed distributions that include it, thereby preventing double counting of variation already represented in the observed dispersion [8–10].

1.3.Measurement distributions

Each population is modeled independently as normal, lognormal, or gamma using arithmetic-scale means and standard deviations. The selected family determines the distribution tails and the third and fourth central moments used in sampling covariance, affecting both point estimates and uncertainty [11–13].

1.4.Previous research

Parametric ROC analysis and analytical measurement uncertainty offer complementary perspectives on diagnostic performance [8–10, 13–16]. Threshold-specific uncertainty analysis connects these perspectives through explicitly defined primitive inputs and dependence assumptions.

Previous tools examined relationships between diagnostic accuracy and measurement uncertainty, diagnostic-measure uncertainty, and Bayesian diagnostic probabilities [17–19]. The present framework uses an implementation that explicitly accounts for covariance among the input estimates. The framework retains within-population mean–standard deviation covariance and dependence induced by a common measurement bias. Its source decomposition follows the covariance form of the law of propagation of uncertainty [17–19].

2. Methods

2.1.Study inputs and diagnostic performance

We modeled test measurements separately in diseased and nondiseased groups, using a normal, lognormal, or gamma distribution for each group. The inputs were each group’s sample size, arithmetic mean, and standard deviation, together with disease prevalence, the specified fixed cutoff,

and the standard uncertainty of a common measurement bias. A result at or above the threshold was classified as positive. The fitted distributions gave Se and Sp; these values and prevalence were then used to calculate the sixteen measures listed in Table 2.

2.2.Sources of uncertainty

We included uncertainty arising from the finite samples, from prevalence when applicable, and from a common measurement bias [6]. The calculation retained the relationship between the estimated mean and standard deviation within each group. The same measurement bias was applied to both groups at the fixed diagnostic threshold: a positive bias increases Se and decreases Sp. Accounting for this shared response is necessary when calculating uncertainty in measures that combine both rates.

The common measurement bias was evaluated separately from independent analytical variation in individual measurements. In the illustrative analysis, approximate population distributions were adjusted by subtracting the mean analytical variance. This adjustment does not establish a common measurement uncertainty component or justify treating independent variation as a common shift. The common component requires separate empirical justification; uncertainty from fitting and variance adjustment was not propagated separately.

2.3.Prevalence

For the illustrative cohort calculation, prevalence was the proportion classified as diseased, with binomial sampling uncertainty. In a case–control study with deliberately fixed group sizes, that proportion need not represent clinical prevalence.

2.4.Uncertainty estimates and intervals

At each specified threshold, uncertainty was propagated using a first-order approximation, yielding source-specific contributions, the combined standard uncertainty, and an approximate pointwise expanded-uncertainty interval at a specified nominal coverage level [6, 20, 21]. Standard uncertainty is expressed on a standard-deviation scale; the interval gives a wider range around the estimate. Intervals depend on the fitted model and may be less reliable near the limits of a measure or when a ratio has a very small denominator. Restricting an interval to its admissible range does not establish its coverage.

2.5.Threshold comparison and implementation

We evaluated performance and uncertainty across a threshold range. For the optimization analysis, the software selected the threshold that maximized each chosen measure above the nondiseased mean. At that selected threshold, it used nonlinear propagation for the common measurement uncertainty. These intervals quantify uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold; they exclude uncertainty associated with threshold estimation and selection. A clinically preferred threshold also depends on the consequences of false-positive and false-negative results.

Calculations were performed in the *DiagAccU*, Version 2.0.0, Wolfram Language notebook. The case-study inputs and settings are reported in Tables 3 and 4. *Appendix A* gives the formulas, assumptions, derivatives, uncertainty calculations, and general calculation procedures. *Appendix B* lists notation and software information.

3. Results

3.1.Illustrative case study

The illustrative analysis uses fasting plasma glucose (FPG) data from 414 participants aged 65–67 years in NHANES 2005–2016 [27]. Reference classification by 2-h oral glucose tolerance test (OGTT)

identified 52 participants with values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). The analysis is unweighted and does not provide a population estimate.

The reference-positive fraction and its binomial standard uncertainty are:

$$v = \frac{52}{414} = 0.125603865, \quad u_v = 0.016287538$$

Approximate lognormal distributions adjusted for analytical variation were obtained by subtracting the mean analytical variance from each maximum-likelihood fitted variance. Table 3 gives the rounded, approximate parameters used in the uncertainty calculations.

Table 3. Rounded population inputs and lognormal summaries for the illustrative example. Concentrations are in mg/dL.

Analysis input or model quantity	Diseased	Nondiseased
Sample size	52	362
Analysis mean (mg/dL)	136.0	101.0
Analysis SD (mg/dL)	36.6	11.0
Model median (mg/dL)	131.327471809	100.406266869
Model skewness	0.826843601	0.328024529
Model kurtosis (not excess)	4.239774961	3.191905805
Log mean	4.877693989	4.609224624
Log SD	0.264427294	0.108589986

The lognormal distributions specified by the rounded arithmetic means and standard deviations are:

$$X_D \sim \text{Lognormal}(4.877693989, 0.264427294)$$

$$X_H \sim \text{Lognormal}(4.609224624, 0.108589986)$$

The standard uncertainty of the common measurement bias, evaluated at the specified fixed cutoff, is modeled as:

$$u_{\text{meas}} = \sqrt{0.812^2 + (0.0119t)^2}$$

This component represents a common shift in measured values. It is distinct from the independent analytical variation addressed by the variance adjustment. Assigning the stated magnitude to a common measurement effect is an illustrative assumption requiring evidence about the common component; total QC variation alone does not establish that component.

At the clinical threshold of 126 mg/dL, the standard uncertainty of the common measurement bias is:

$$u_{\text{meas}}(126) = 1.705152298 \text{ mg/dL}$$

Table 4. Calculation settings for the illustrative example.

Setting	Value and scope
Population families	Lognormal-lognormal; alternative distribution families are compared in Figure 9.
Sample sizes	$n_D = 52$; $n_H = 362$
Arithmetic means	$m_D = 136.0$; $m_H = 101.0$ mg/dL
Arithmetic SDs	$s_D = 36.6$; $s_H = 11.0$ mg/dL
Threshold	$t = 126$ mg/dL for Figures 1, 4, 9, 10; 91–173 mg/dL for Figures 3, 5–8
Prevalence	$v = 52/414$; varied from 0.001 to 0.999 only in Figure 4
Measurement uncertainty coefficients	$b_0 = 0.812$ mg/dL; $b_1 = 0.0119$
Prevalence sampling	Enabled.
Sampling degrees of freedom	$\nu_D = 51$; $\nu_H = 361$; $\nu_v = 413$
Measurement-source degrees of freedom	$\nu_{\text{meas}} = \infty$
Intervals	Nominal two-sided 95%; Coverage factor derived from Student's t distribution using effective degrees of freedom calculated from the contributions of the independent uncertainty sources.; admissible-range restriction

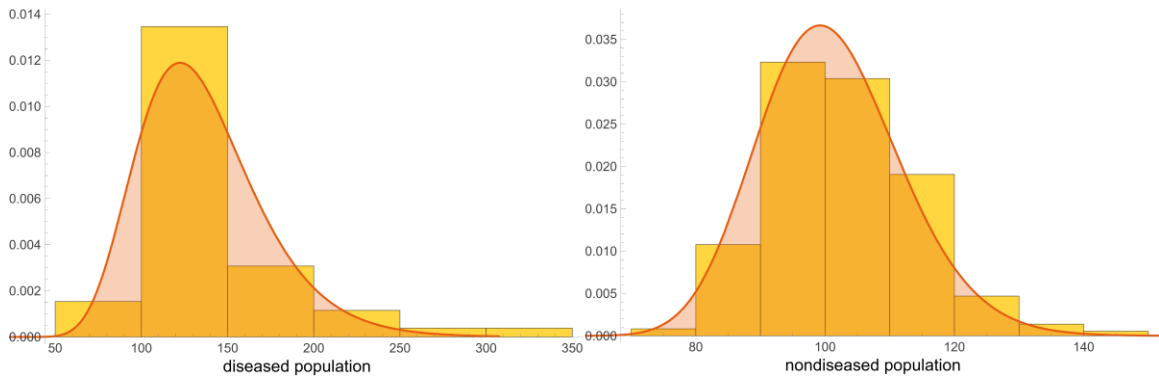


Figure 2. FPG histograms (yellow) and approximate lognormal distributions (orange) for 414 NHANES participants aged 65–67 years: 52 with 2-h OGTT values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). FPG is in mg/dL.

Figure 1 illustrates classification at 126 mg/dL, with FPG values at or above this threshold classified as positive.

3.2. Program application

The measures and their uncertainties are evaluated over 91–173 mg/dL using the settings in Table 4. Figure 4 examines prevalence effects at a fixed threshold.

3.3. Threshold-dependent profiles

Increasing the threshold reduces Se and increases Sp and, generally, PPV. NPV exhibits a shallow maximum before declining, whereas ACC peaks near 130 mg/dL. The measures therefore favor different thresholds (Figure 3).

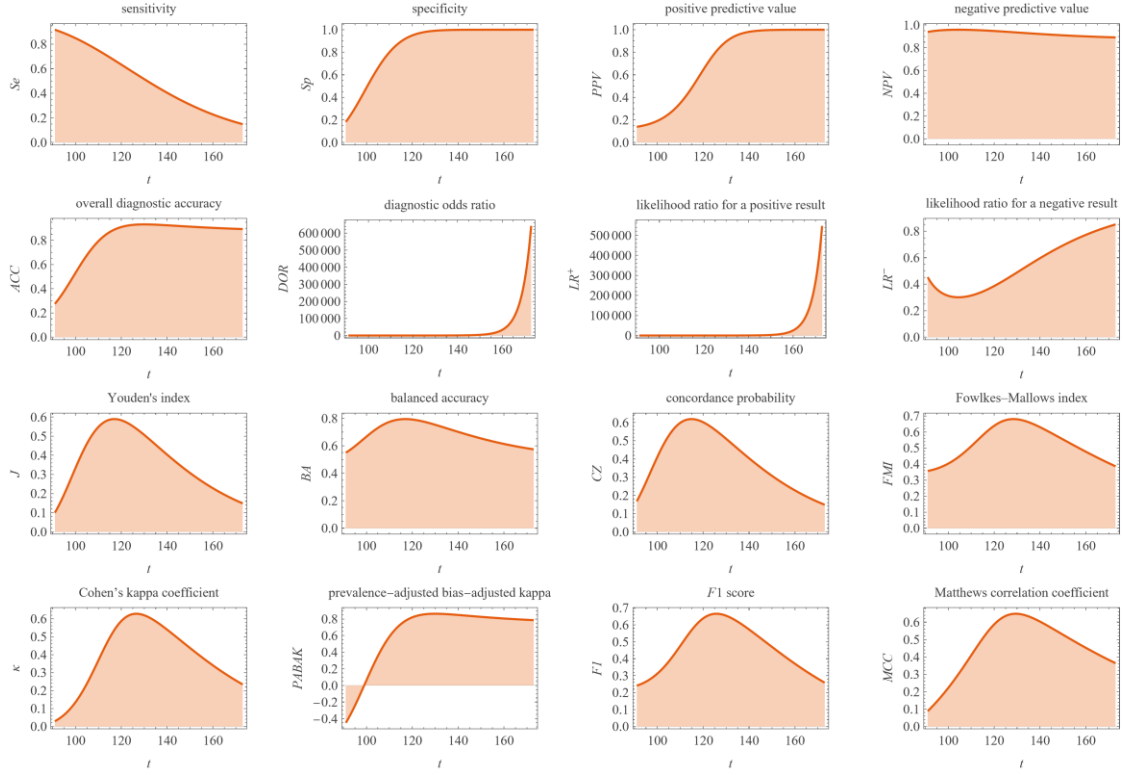


Figure 3. Profiles of the 16 diagnostic accuracy measures over thresholds of 91–173 mg/dL, using the population inputs and measurement-uncertainty settings in Tables 3 and 4.

3.4. Prevalence-dependent profiles

At 126 mg/dL, increasing prevalence raises PPV and lowers NPV. ACC and PABAK decrease, whereas FMI and F_1 increase. κ and MCC attain interior maxima (Figure 4).

The profiles show point estimates with fixed population distributions; prevalence-independent measures are omitted.

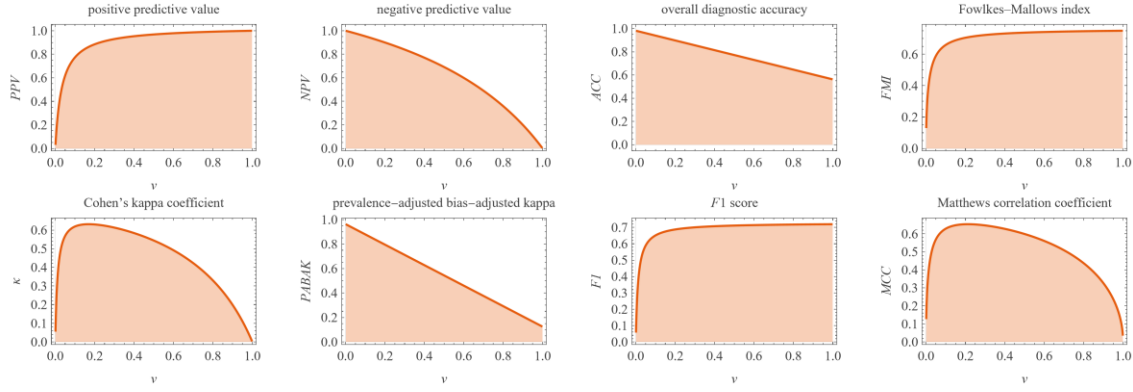


Figure 4. Profiles of eight diagnostic accuracy measures over prevalence values of 0.001–0.999, with the FPG threshold fixed at 126 mg/dL and the population distributions held constant.

3.5. Relations among measures

Figure 5 relates selected pairs of measures evaluated at a common threshold. Direction reversals indicate that one value of a measure may correspond to multiple values of another.

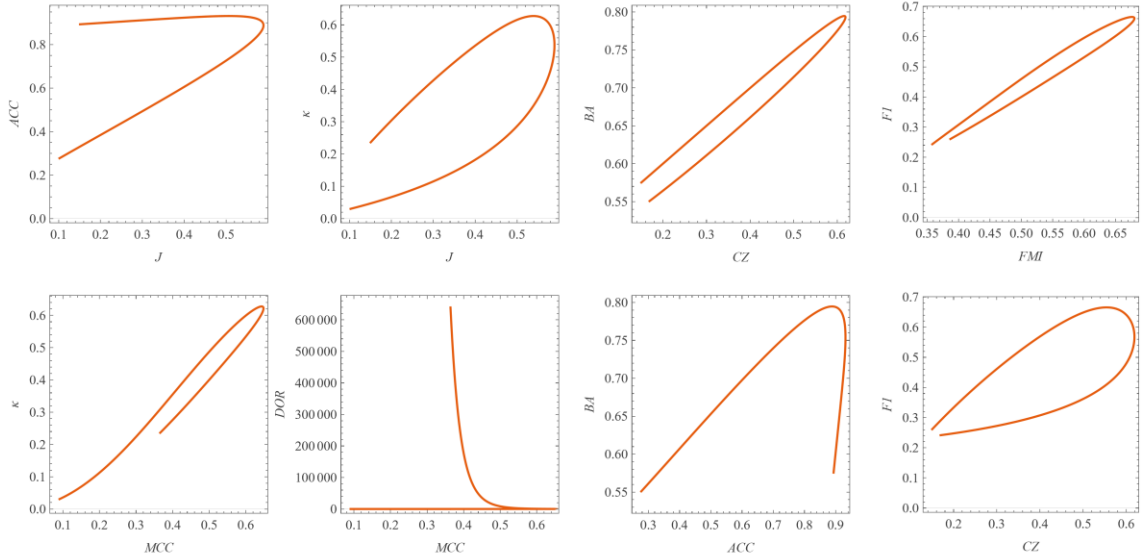


Figure 5. Relations among diagnostic accuracy measures over thresholds of 91–173 mg/dL. Each curve pairs measures evaluated at a common threshold. CZ denotes the Se–Sp product.

3.6.Uncertainty

Figure 6 presents total sampling, measurement, and combined standard uncertainties, with covariance retained in the propagation.

At 126 mg/dL, PPV has total sampling and measurement standard uncertainties of 0.0441 and 0.0406, respectively, and a combined standard uncertainty of 0.0600.

Sampling uncertainty dominates many measures near 126 mg/dL, whereas measurement uncertainty predominates for Sp and LR⁺. Cancellation between Se and Sp responses can reduce the first-order measurement contribution.

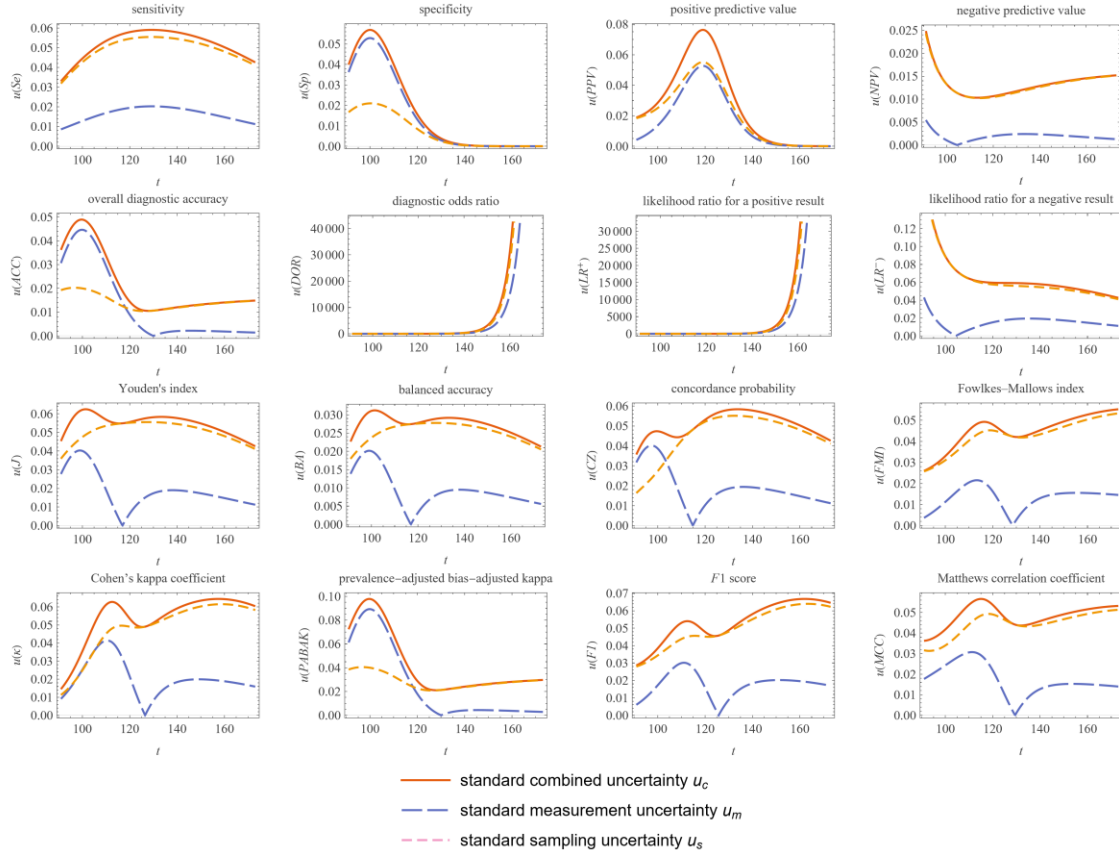


Figure 6. Combined (solid orange), measurement (long-dashed blue), and total sampling (short-dashed yellow-orange) standard uncertainties across thresholds. Sampling includes both populations and the enabled prevalence component; covariance is retained within each source. The measurement component shown in the software output is the common measurement contribution defined here.

3.7. Relative uncertainty

Relative standard uncertainty is the standard uncertainty divided by the absolute measure value. It can become very large near zero and is undefined at zero; neither feature independently indicates a change in diagnostic performance.

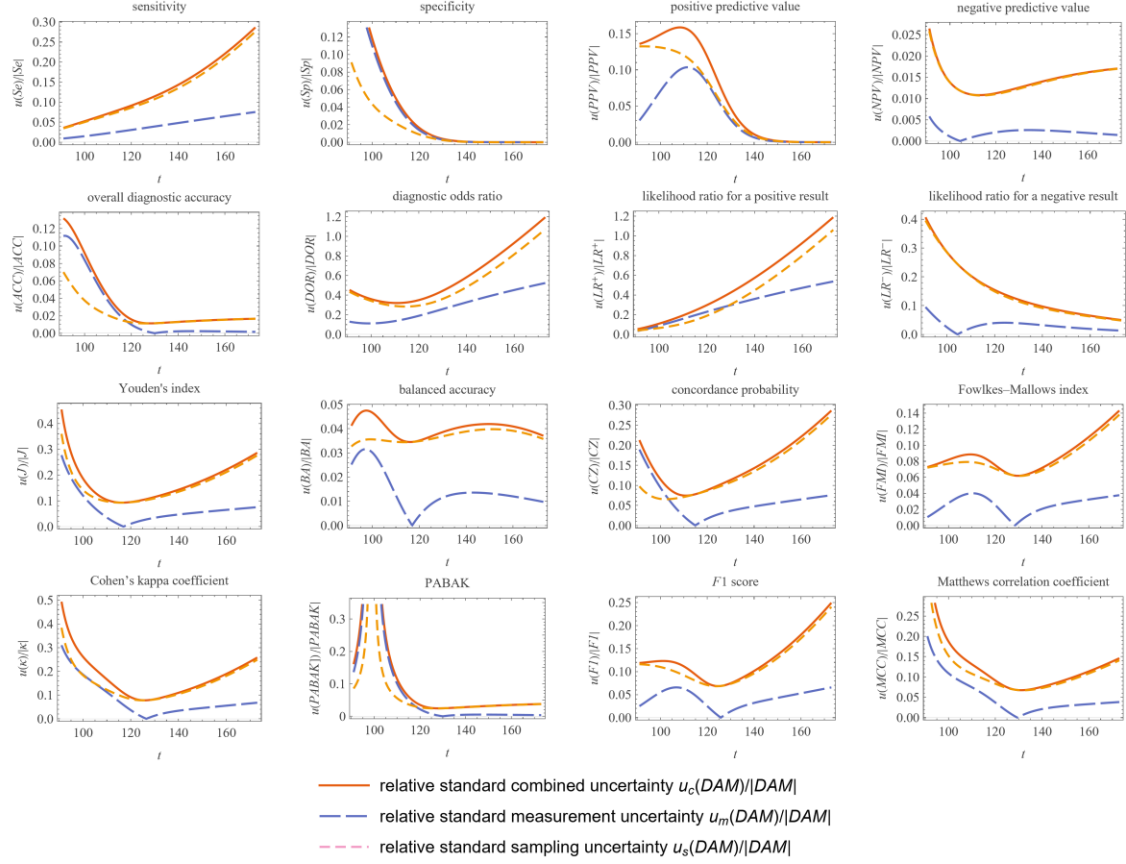


Figure 7. Relative standard uncertainties of the diagnostic accuracy measures, using the component styles in Figure 6. Relative uncertainty is undefined at zero measure values.

3.8. Pointwise uncertainty intervals

Figure 8 shows approximate pointwise 95% expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions.. Interval limits are restricted to each measure's admissible range. The nominal 95% level refers to each fixed threshold individually; these intervals do not constitute simultaneous confidence bands across thresholds.

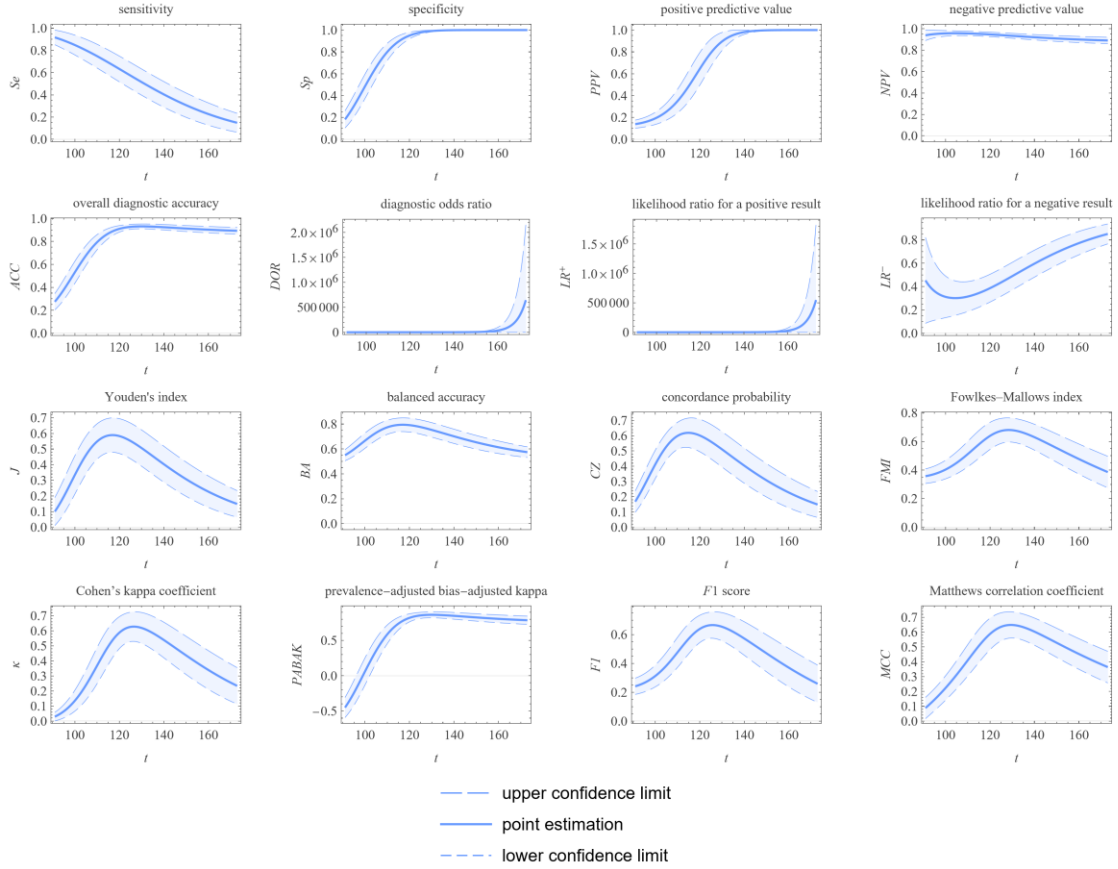


Figure 8. Point estimates (curves) and approximate pointwise 95% expanded-uncertainty intervals (shading and dashed interval limits) across thresholds.

3.9. Estimates and intervals at the clinical threshold

At 126 mg/dL, PPV is 0.8155, with standard uncertainty 0.0600 and an approximate 95% expanded-uncertainty interval of 0.6980–0.9331. NPV is 0.9398, with standard uncertainty 0.0113 and an approximate 95% expanded-uncertainty interval of 0.9176–0.9620. Figure 9 compares distribution families; Figure 10 presents estimates and intervals for all 16 measures under the lognormal model.

point estimates of diagnostic accuracy measures																	
prevalence of disease $\nu = 0.126$																	
measurement distribution		Se	Sp	ACC	PPV	NPV	DOR	LR ⁺	LR ⁻	J	BA	CZ	FMI	κ	PABAK	F ₁	MCC
diseased	nondiseased																
normal	normal	0.608	0.988	0.941	0.883	0.946	132.880	52.742	0.397	0.596	0.798	0.601	0.733	0.688	0.881	0.720	0.703
	lognormal	0.608	0.982	0.935	0.827	0.946	83.244	33.268	0.400	0.589	0.795	0.597	0.709	0.665	0.869	0.701	0.675
	gamma	0.608	0.984	0.937	0.845	0.946	95.146	37.937	0.399	0.592	0.796	0.598	0.717	0.673	0.873	0.707	0.684
lognormal	normal	0.562	0.988	0.935	0.875	0.940	110.190	48.799	0.443	0.551	0.775	0.556	0.701	0.650	0.870	0.685	0.670
	lognormal	0.562	0.982	0.929	0.816	0.940	69.026	30.780	0.446	0.544	0.772	0.552	0.677	0.627	0.858	0.666	0.641
	gamma	0.562	0.984	0.931	0.834	0.940	78.895	35.101	0.445	0.546	0.773	0.553	0.685	0.635	0.862	0.672	0.650
gamma	normal	0.575	0.988	0.937	0.878	0.942	116.210	49.931	0.430	0.564	0.782	0.569	0.711	0.661	0.873	0.695	0.680
	lognormal	0.575	0.982	0.931	0.819	0.941	72.797	31.494	0.433	0.557	0.779	0.565	0.686	0.638	0.861	0.676	0.651
	gamma	0.575	0.984	0.933	0.838	0.942	83.206	35.915	0.432	0.559	0.780	0.566	0.694	0.646	0.865	0.682	0.660

Figure 9. Diagnostic accuracy estimates at 126 mg/dL for nine combinations of normal, lognormal, and gamma population distributions, with arithmetic means, standard deviations, sample sizes, and prevalence held fixed.

95.00% confidence intervals			
prevalence of disease $\nu = 0.126$			
measure	point estimate	lower confidence limit	upper confidence limit
Se	0.562	0.445	0.680
Sp	0.982	0.968	0.996
ACC	0.929	0.908	0.950
PPV	0.816	0.698	0.933
NPV	0.940	0.918	0.962
DOR	69.026	14.805	123.250
LR ⁺	30.780	8.439	53.121
LR ⁻	0.446	0.327	0.564
J	0.544	0.429	0.659
BA	0.772	0.715	0.829
CZ	0.552	0.438	0.666
FMI	0.677	0.591	0.764
κ	0.627	0.530	0.725
PABAK	0.858	0.816	0.900
F ₁	0.666	0.574	0.757
MCC	0.641	0.548	0.734

Figure 10. Point estimates and approximate pointwise 95% expanded-uncertainty intervals for the 16 diagnostic accuracy measures at 126 mg/dL under the lognormal model.

3.10. Optimal diagnostic thresholds

CZ is maximized at 114.859 mg/dL, J and BA at 116.948 mg/dL, and ACC and PABAK at 130.011 mg/dL. Although the first-order threshold derivative vanishes at a smooth optimum, nonlinear propagation retains a measurement contribution. For ACC, the measurement and combined standard uncertainties are 0.000568 and 0.010659, respectively. The intervals are conditional on the estimated optimal threshold and exclude uncertainty from repeated threshold selection.

95.00% confidence intervals				
prevalence of disease $\nu = 0.126$				
measure	optimal threshold t	point estimate	lower confidence limit	upper confidence limit
ACC	130.011	0.932	0.911	0.953
J	116.948	0.589	0.479	0.700
BA	116.948	0.795	0.740	0.850
CZ	114.859	0.619	0.522	0.716
FMI	128.416	0.680	0.596	0.765
κ	126.529	0.628	0.529	0.726
PABAK	130.011	0.863	0.821	0.905
F_1	125.753	0.666	0.574	0.757
MCC	129.421	0.648	0.560	0.737

Figure 11. Optimal thresholds and corresponding measure values subject to $t > m_H$. Combined and measurement standard uncertainties and approximate 95% intervals of the measures.

4. Discussion

4.1. Principal findings

The framework propagates uncertainty for sixteen measures across nine population-family pairs. The PPV and NPV results illustrate the joint effects of mean–standard deviation covariance and a common measurement bias.

Maximizing a point estimate need not minimize combined or relative uncertainty. At a smooth optimum, the first-order threshold derivative may vanish while nonlinear propagation of uncertainty in the measurement bias still contributes to uncertainty. Neither derivative cancellation nor a high estimate establishes clinical preference.

4.2. Relations between measures

The sixteen outputs depend on Se, Sp, and prevalence, so their first-order covariance has rank at most three. They provide complementary interpretations rather than sixteen independent assessments. The identities relating BA to J and PABAK to ACC explain their common maximizing thresholds and proportional uncertainties.

4.3. Uncertainty decomposition

Accounting for covariance among inputs from the same uncertainty source affects the propagated variance. The measurement contribution combines both tail derivatives before squaring, accounting for the single measurement-uncertainty input.

Excluding prevalence uncertainty changes the combined standard uncertainty of PPV from 0.059962 to 0.055657 and that of NPV from 0.011265 to 0.007517, with unchanged central estimates. The

sampling interpretation determines whether this source is included; the smaller uncertainty is not a selection criterion.

4.4. Additional sources of uncertainty and bias

Reference classification: Group labels are treated as fixed. OGTT misclassification, reference-standard uncertainty, verification bias, and incomplete disease ascertainment are excluded [28].

Distributional specification: Normal, lognormal, and gamma models may inadequately represent mixtures, truncation, censoring, or heavy tails. Checks across the nine family pairs verify executable support, not fit to participant data [11–13, 29].

Prevalence and transportability: Modeled-cohort uncertainty assumes a binomial count ratio independent of the fitted summaries. Fixed strata, survey sampling, and deployment populations may require different prevalence inputs and additional covariance terms.

Behavior near parameter limits: First-order intervals may perform poorly near zero denominators or parameter limits. Transformed, likelihood-based, joint Monte Carlo, and bootstrap alternatives require separate assumptions and validation; admissible-range restriction alone is insufficient [30–33].

Sampling approximations: The sample-mean variance is exact under the stated independent-sample model with finite variance; covariance terms involving the sample standard deviation are approximated to first order. Normality does not make the propagated DAM uncertainty or interval exact.

Threshold selection: Selecting a threshold from fitted inputs introduces uncertainty and dependence omitted by a conditional interval. Resampling should repeat model estimation and threshold selection in each replicate [34–36].

Transfer of the measurement-uncertainty model: Coefficients b_0 and b_1 are treated as known and shared by both populations. Propagating their estimation covariance, matrix or instrument changes, or correlations with fitted summaries requires an expanded input model.

Dependence: The model includes within-population mean–standard deviation covariance but excludes cross-population covariance, clustering, repeated measurements, and dependence between prevalence and fitted summaries. The common measurement bias is assumed to be independent of fitted summaries.

Decision rule and intervals: Reversing the positive-test orientation requires redefining both tails. Pointwise intervals across measures and thresholds are not simultaneous intervals; admissible-range restriction does not establish nominal coverage.

4.5. Implementation strengths

Common parameterization: Arithmetic means and standard deviations retain the same interpretation across all three population families.

Source accounting: Uncertainty in the estimated population means and standard deviations, prevalence, and the common measurement bias is propagated to each measure, except where a contribution is identically zero or explicitly excluded.

Scalar–joint agreement: The joint covariance diagonal reproduces scalar combined variances to numerical precision, and source covariance matrices sum to the total.

Reproducibility: The source and calculations Wolfram notebooks document the calculations.

4.6. Contribution

The contribution is a consistent, reproducible sixteen-measure implementation of established uncertainty methods under explicit assumptions.

4.7. Reporting recommendations

- a) Specify the positive-test orientation, population families, arithmetic summaries, sample sizes, and prevalence interpretation.
- b) Report point estimates, source-specific and combined standard uncertainties, effective degrees of freedom, coverage factors, interval levels, and any endpoint restriction.
- c) Distinguish fixed-threshold intervals from intervals conditional on a fitted optimum. Assess nonlinear and selection effects through a specified joint simulation or resampling analysis where needed.
- d) Specify the standard-uncertainty model for the common measurement bias, coefficient values and units, and treatment of coefficient uncertainty and finite measurement degrees of freedom.

4.8. Limitations and future work

The calculations are based on rounded illustrative inputs; Section 3.1 and Figure 2 describe the reconstruction of the empirical NHANES distributions and the corresponding adjusted distribution fits. Interpretation is limited by first-order uncertainty propagation, effective degrees of freedom derived from source-specific variance contributions, measurement-uncertainty coefficients treated as known, and intervals conditional on the estimated optimal threshold. Future work should evaluate interval coverage through simulation, propagate uncertainty and covariance in the measurement-uncertainty coefficients, incorporate survey design and reference-standard uncertainty, and repeat model estimation and threshold selection in each bootstrap replicate. [36–42].

5. Conclusion

The framework quantifies uncertainty for sixteen diagnostic accuracy measures under independently specified normal, lognormal, or gamma population models, retaining within-population mean–standard deviation covariance and dependence.

Sampling contributes substantially, and cancellation of the common measurement contribution varies by measure; these results do not establish universal dominance across measures or thresholds.

The joint covariance represents dependence among all sixteen outputs. Checks across all nine family pairs, scalar–joint agreement, and source-matrix addition support implementation consistency.

Model-based maxima permit comparison of diagnostic objectives within the optimization domain. Clinical threshold selection additionally requires evidence on consequences, transportability, and uncertainty from estimation and selection.

6. Supplemental material

The following supplemental files are available for download as a ZIP archive at:
<https://www.hcsl.com/Supplements/SDAMUR.zip> (accessed on September 21, 2026):

- a) Supplemental File I:
DiagAccU.nb: The program as a Wolfram Mathematica Notebook.
- b) Supplemental File II:
DiagAccUCalculations.nb: The calculations for the estimation of the DAMs and their combined standard uncertainty in a Wolfram Mathematica Notebook.

- c) Supplemental File III:
DiagAccUInterface.pdf: A brief documentation of the interface of the program.

7. Declarations

Author Contributions: Conceptualization: RCA and TC; methodology: RCA and ATH; software: RCA, TC and ATH; validation: TC; formal analysis: RCA and ATH; investigation: TC; resources: ATH; data curation: TC; writing—original draft preparation: RCA; writing—review and editing: ATH; visualization: RCA and TC; supervision: ATH; project administration: RCA. All authors have read and agreed to the published version of the manuscript.

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Appendix A. Mathematical and statistical calculations

This appendix presents the mathematical and statistical calculations from input specification to expanded-uncertainty intervals. Section A.5 provides the complete derivative table, and Section A.7 gives the joint covariance formulation.

Variance is the square of the corresponding standard uncertainty. Covariance describes how two uncertain quantities vary together. A partial derivative describes the local rate of change of an output with respect to one input, with all other inputs held fixed. A gradient collects the partial derivatives for one output, and a Jacobian collects them for several outputs. First-order uncertainty propagation is based on a local linear approximation.

A.1. Input specification and ordering

The population labels D and H denote diseased and nondiseased populations, respectively. Prevalence is denoted v , and the intermediate arguments a and b denote Se and Sp. A generic diagnostic accuracy measure is denoted F . The standard uncertainties u_c , u_s , and u_m , evaluated for a specified measure, denote combined, total sampling, and measurement uncertainty, respectively. The measurement component represents uncertainty in a common measurement bias. The specified diagnostic cutoff is fixed; the transformed threshold represents uncertainty in the common measurement bias. The population-parameter and prevalence sampling components are denoted u_{par} and u_{prev} . Superscript T denotes transpose. The measure is a function of the primitive inputs, $F = g(\mathbf{x})$.

Scalar variables are italic; measure abbreviations, named operators, and descriptive subscripts are upright. Vectors and matrices are bold. The primitive-input Jacobian is denoted $\mathbf{M}$, and the four source-specific variance contributions are denoted V_D , V_H , V_v , and V_{meas} . These correspond to the Jacobian $\mathbf{J}$ and the variance contributions V_D , V_H , V_v , and V_{meas} , respectively, in Appendix B. The symbol J denotes Youden's index throughout this appendix.

The primitive input vector comprises the four fitted arithmetic summaries, prevalence, and the transformed threshold induced by the common measurement bias. The intermediate vector comprises the three arguments of each diagnostic accuracy measure. The sixth coordinate is $\tau = t - \Delta$, evaluated at $\tau = t$ for the nominal inputs. Its variance is $u_{\text{meas}}^2(t)$. Derivatives with respect to the displayed cutoff argument t are used as transformed-threshold sensitivity coefficients at $\tau = t$. Derivatives with respect to Δ reverse both tail-derivative signs and therefore give the same covariance contribution.

$$\mathbf{x} = \begin{pmatrix} m_D \\ s_D \\ m_H \\ s_H \\ v \\ \tau \end{pmatrix} \in \mathbb{R}^6, \quad (\text{A.1})$$

$$\mathbf{z} = \begin{pmatrix} a \\ b \\ v \end{pmatrix} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ v \end{pmatrix} \in \mathbb{R}^3.$$

The primitive and intermediate vectors have six and three components, respectively. All vectors are column vectors. The aliases $a = \text{Se}$ and $b = \text{Sp}$ are used in the derivative formulas; v denotes prevalence throughout.

$$\mathbf{m} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ \text{ACC} \\ \text{PPV} \\ \text{NPV} \\ \text{DOR} \\ \text{LR}^+ \\ \text{LR}^- \\ J \\ \text{BA} \\ \text{CZ} \\ \text{FMI} \\ \kappa \\ \text{PABAK} \\ F_1 \\ \text{MCC} \end{pmatrix} \in \mathbb{R}^{16}. \quad (\text{A.2})$$

Equation (A.2) specifies the output order. Row i of each output Jacobian corresponds to measure i , and element (i, j) of the output covariance matrix is the covariance between measures i and j .

Observations are assumed to be independent within each population, and the population samples are assumed to be independent of one another. The supplied standard deviations are assumed to be calculated from Bessel-corrected sample variances. Prevalence and the common measurement bias are modeled as mutually independent inputs, each independent of the fitted population summaries. These assumptions specify which elements of the input covariance matrix are zero. For other study designs, the covariance matrix must reflect the corresponding dependence structure.

A.2. Distribution fitting and calculation of the two classification rates

For population $P \in \{D, H\}$, the normal, lognormal, and gamma families are parameterized by the arithmetic mean μ_P and standard deviation σ_P . Lognormal and gamma models require positive means, and positive thresholds are required whenever either population has positive support. The gamma parameters ξ and θ denote shape and scale, respectively.

$$\begin{aligned} X_N &\sim \mathcal{N}(\mu, \sigma^2). \\ \sigma_L^2 &= \log\left(1 + \left(\frac{\sigma}{\mu}\right)^2\right), \\ \mu_L &= \log(\mu) - \frac{\sigma_L^2}{2}. \\ X_L &\sim \text{Lognormal}(\mu_L, \sigma_L). \\ \xi &= \left(\frac{\mu}{\sigma}\right)^2, \quad \theta = \frac{\sigma^2}{\mu}, \\ X_G &\sim \text{Gamma}(\xi, \theta). \end{aligned}$$

The selected distribution family is parameterized using the supplied arithmetic mean and standard deviation. For the lognormal family, the transformed parameters describe the logarithm of the measurement. For the gamma family, the shape and scale are obtained from the two arithmetic summaries. Agreement with the arithmetic summaries does not establish adequate fit in the distribution tails.

For a positive-above-threshold test, Se is the probability in the diseased upper tail and Sp is the probability in the nondiseased lower tail. A cumulative distribution function gives the probability below a threshold. Thus Se is one minus the diseased cumulative probability, whereas Sp is the nondiseased cumulative probability.

The class-conditional cumulative distribution functions are denoted F_D and F_H . Under the positive-above-threshold rule, Se and Sp are defined as follows:

$$\text{Se}(t) = \Pr(X_D \geq t).$$

$$\text{Sp}(t) = \Pr(X_H < t).$$

$$\text{Se}(t) = 1 - F_D(t), \quad \text{Sp}(t) = F_H(t).$$

For continuous distributions, including all three families used here, the probability of observing the threshold exactly is zero. The distinction between a strict and a non-strict inequality therefore does not alter the model probabilities.

A.3. Calculation of diagnostic accuracy measures

Tables A1 and A2 define the remaining measures. The symbols v , q , p_o , and p_e denote prevalence, predicted-positive probability, observed agreement, and chance agreement, respectively:

$$q = v \text{Se} + (1 - v) (1 - \text{Sp}).$$

$$p_o = v \text{Se} + (1 - v) \text{Sp}.$$

$$p_e = q v + (1 - q) (1 - v).$$

Table A1. Mathematical definitions of Se, Sp, PPV, NPV, ACC, LR⁺, LR[−], and balance measures.

Measure	Count definition	Definition using Se, Sp, v
Se	$\frac{\text{TP}}{\text{TP} + \text{FN}}$	Se
Sp	$\frac{\text{TN}}{\text{TN} + \text{FP}}$	Sp
PPV	$\frac{\text{TP}}{\text{TP} + \text{FP}}$	$\frac{v\text{Se}}{q}$
NPV	$\frac{\text{TN}}{\text{TN} + \text{FN}}$	$\frac{(1 - v)\text{Sp}}{1 - q}$
ACC	$\frac{\text{TP} + \text{TN}}{N}$	$v\text{Se} + (1 - v)\text{Sp}$
DOR	$\frac{\text{TPTN}}{\text{FPFN}}$	$\frac{\text{Se Sp}}{(1 - \text{Se})(1 - \text{Sp})}$
LR ⁺	$\frac{\frac{\text{TP}}{R}}{\frac{\text{FP}}{N - R}}$	$\frac{\text{Se}}{1 - \text{Sp}}$
LR [−]	$\frac{\frac{\text{FN}}{R}}{\frac{\text{TN}}{N - R}}$	$\frac{1 - \text{Se}}{\text{Sp}}$
J	$\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R} - 1$	$\text{Se} + \text{Sp} - 1$
BA	$\frac{\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R}}{2}$	$\frac{\text{Se} + \text{Sp}}{2}$
CZ	$\frac{\text{TPTN}}{R(N - R)}$	Se Sp

Table A2. Mathematical definitions of association and agreement measures.

Measure	Count definition	Definition using Se, Sp, v
FMI	$\frac{TP}{\sqrt{RN_+}}$	$\sqrt{Se PPV}$
κ	$\frac{N(TP + TN) - RN_+ - (N - R)(N - N_+)}{N^2 - RN_+ - (N - R)(N - N_+)}$	$\frac{p_o - p_e}{1 - p_e}$
PABAK	$2 \frac{TP + TN}{N} - 1$	$2p_o - 1$
F_1	$\frac{2TP}{2TP + FP + FN}$	$\frac{2vSe}{v + q}$
MCC	$\frac{TPTN - FPFN}{\sqrt{RN_+(N - R)(N - N_+)}}$	$\frac{\sqrt{v(1 - v)}(Se + Sp - 1)}{\sqrt{q(1 - q)}}$

In Tables A1 and A2, $N = TP + FN + FP + TN$, $R = TP + FN$, and $N_+ = TP + FP$ denote total, disease-positive marginal, and test-positive marginal counts, respectively. Dividing counts by N gives the equivalent probability definitions. Ratios and their derivatives require nonzero denominators.

Classification of measures

Table A3. Overlapping classification of the sixteen measures.

Category	Measures
Agreement	κ , PABAK
Association and F-type indices	FMI, F_1 , MCC
Balance	J , BA, CZ
Bounded	Se, Sp, ACC, PPV, NPV, J , BA, CZ, FMI, κ , PABAK, F_1 , MCC
Ratio-type	DOR, LR^+ , LR^-
Prevalence invariant	Se, Sp, DOR, LR^+ , LR^- , J , BA, CZ
Prevalence dependent	ACC, PPV, NPV, FMI, κ , PABAK, F_1 , MCC

The classifications overlap: agreement measures and F-type indices may depend on prevalence. Boundedness implies neither low uncertainty nor robustness to class imbalance.

The predicted-positive probability equals the sum of the probabilities of true-positive and false-positive outcomes in the target population. PPV is the ratio of the true-positive probability to this sum. Similarly, NPV is the ratio of the true-negative probability to the sum of the true-negative and false-negative probabilities. Agreement and association measures are also calculated from these outcome probabilities.

A.4. Construction of the input covariance matrix

The input covariance matrix includes the variances of the estimated means and standard deviations for both populations, together with the prevalence variance and the variance of the common measurement bias. Within each population, the covariance between the estimated mean and standard deviation is retained. All remaining covariances are zero under the stated independence assumptions [6, 44, 45].

A.4.1. Sampling uncertainty within each population

The centered sampling deviations are defined as $\delta_{\bar{X}} = \bar{X} - E[\bar{X}]$ and $\delta_S = S - E[S]$, where expectations are taken over repeated samples from the same population.

$$\begin{aligned}\Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}} \\ \delta_S \end{pmatrix} (\delta_{\bar{X}} \quad \delta_S) \right), \\ \Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}}^2 & \delta_{\bar{X}} \delta_S \\ \delta_S \delta_{\bar{X}} & \delta_S^2 \end{pmatrix} \right), \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} A & B \\ B & C \end{pmatrix}.\end{aligned}\tag{A.3}$$

The covariance matrix is the expectation of the outer product of the centered estimator vector with its transpose. Expectations are evaluated for each matrix element. The off-diagonal elements are equal by symmetry of covariance.

$$\begin{aligned}\Sigma_{\bar{X},S^2} &= \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{n} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{n} \end{pmatrix}, \\ \mathbf{L} &= \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2\sigma} \end{pmatrix}.\end{aligned}\tag{A.4}$$

The transformation starts from the covariance of the sample mean and Bessel-corrected sample variance. Its Jacobian $\mathbf{L}$ leaves the mean unchanged and maps variance to its positive square root.

$$\mathbf{L} \Sigma_{\bar{X},S^2} = \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{2n\sigma} \end{pmatrix}.\tag{A.5}$$

Left multiplication by $\mathbf{L}$ divides the second row by twice the population standard deviation.

$$\begin{aligned}\Sigma_{\bar{X},S} &\approx \mathbf{L} \Sigma_{\bar{X},S^2} \mathbf{L}^T, \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{2n\sigma} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{4n\sigma^2} \end{pmatrix}.\end{aligned}\tag{A.6}$$

Right multiplication by the transpose of $\mathbf{L}$ scales the second column. Together, the two multiplications divide the mean–variance covariance by twice the population standard deviation and the variance of the sample variance by four times the population variance, yielding the first-order mean–standard deviation covariance matrix.

$$\begin{aligned}A &= \frac{\sigma^2}{n}, \\ B &= \frac{\mu_3}{2n\sigma}, \\ C &= \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{4n\sigma^2}.\end{aligned}\tag{A.7}$$

A is the exact variance of the sample mean. B and C are first-order approximations to the mean–standard deviation covariance and the variance of the sample standard deviation, respectively. Their dependence on the

third and fourth central moments makes B and C distribution dependent. For a normal population, B is zero; it is generally nonzero for the lognormal and gamma families [45].

The third and fourth central moments reflect distributional asymmetry and the contribution of large deviations from the mean, respectively. The family-specific expressions below are evaluated using the estimated arithmetic means and standard deviations. The reported uncertainty is therefore obtained by substituting parameter estimates into these expressions, even when the underlying variance identities are exact. For the lognormal expressions, $c = 1 + \left(\frac{\sigma}{\mu}\right)^2$.

$$\begin{aligned} \text{Normal: } \mu_3 &= 0, \quad \mu_4 = 3\sigma^4; \\ \text{Lognormal: } \mu_3 &= \frac{(c+2)\sigma^4}{\mu}, \quad \mu_4 = (c^4 + 2c^3 + 3c^2 - 3)\sigma^4; \\ \text{Gamma: } \mu_3 &= \frac{2\sigma^4}{\mu}, \quad \mu_4 = 3\sigma^4 + \frac{6\sigma^6}{\mu^2}. \end{aligned}$$

$$\begin{aligned} \Sigma_D &= \begin{pmatrix} A_D & B_D \\ B_D & C_D \end{pmatrix}, \\ \Sigma_H &= \begin{pmatrix} A_H & B_H \\ B_H & C_H \end{pmatrix}. \end{aligned} \tag{A.8}$$

For each population, the sample arithmetic mean precedes the sample standard deviation in the input vector. The corresponding covariance matrices are symmetric.

$$\Sigma_{\text{par}} = \begin{pmatrix} A_D & B_D & 0 & 0 \\ B_D & C_D & 0 & 0 \\ 0 & 0 & A_H & B_H \\ 0 & 0 & B_H & C_H \end{pmatrix}. \tag{A.9}$$

The matrix Σ_{par} contains the variances and covariances of the four fitted population summaries. The upper-left and lower-right submatrices correspond to the diseased and nondiseased populations, respectively. Covariances between estimators from different populations are zero under the assumption of independent samples.

$$\Sigma_x = \begin{pmatrix} A_D & B_D & 0 & 0 & 0 & 0 \\ B_D & C_D & 0 & 0 & 0 & 0 \\ 0 & 0 & A_H & B_H & 0 & 0 \\ 0 & 0 & B_H & C_H & 0 & 0 \\ 0 & 0 & 0 & 0 & u_v^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & u_{\text{meas}}^2 \end{pmatrix}. \tag{A.10}$$

Including prevalence and the transformed threshold alongside the four fitted population summaries extends the input covariance matrix from 4×4 to 6×6 . Their variances occupy matrix elements (5,5) and (6,6), respectively. All additional off-diagonal elements are zero under the assumed independence of these inputs from the fitted population summaries and from one another.

A.4.2. Prevalence uncertainty

Under the assumed cohort sampling model, prevalence and its standard uncertainty are estimated as follows:

$$\begin{aligned} v &= \frac{n_D}{n_D + n_H}, \\ u_v &= \sqrt{\frac{v(1-v)}{n_D + n_H}}. \end{aligned}$$

A.4.3. Common measurement uncertainty

The standard uncertainty of the common measurement bias, evaluated at the fixed nominal cutoff, is specified by either a linear model, restricted to nonnegative values, or a root-sum-of-squares model. The coefficient b_0 is the constant component and has the measurement unit; b_1 is the dimensionless proportional coefficient. Both coefficients are treated as known. Gauss–Hermite quadrature is a separate procedure for propagating common measurement uncertainty at an estimated optimal threshold.

$$\begin{aligned}
 u_{\text{meas}} &= b_0 + b_1 t \\
 &\text{or} \\
 u_{\text{meas}} &= \sqrt{b_0^2 + (b_1 t)^2}. \\
 \mathbf{h}_{\text{meas}} &= \begin{pmatrix} \frac{\partial \text{Se}}{\partial t} \\ \frac{\partial \text{Sp}}{\partial t} \end{pmatrix} = \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix}.
 \end{aligned} \tag{A.11}$$

The transformed-threshold derivative of Se is nonpositive, whereas that of Sp is nonnegative. The 2×1 sensitivity vector and its transpose determine the covariance contribution from the common measurement bias.

$$\begin{aligned}
 \Sigma_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \mathbf{h}_{\text{meas}} \mathbf{h}_{\text{meas}}^T, \\
 \Sigma_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix} \begin{pmatrix} -f_D(t) & f_H(t) \end{pmatrix}.
 \end{aligned} \tag{A.12}$$

$$\begin{aligned}
 \Sigma_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} (-f_D(t))(-f_D(t)) & (-f_D(t))f_H(t) \\ f_H(t)(-f_D(t)) & f_H(t)f_H(t) \end{pmatrix}, \\
 \Sigma_{\text{Se,Sp;meas}} &= u_{\text{meas}}^2 \begin{pmatrix} f_D(t)^2 & -f_D(t)f_H(t) \\ -f_D(t)f_H(t) & f_H(t)^2 \end{pmatrix}.
 \end{aligned} \tag{A.13}$$

The diagonal elements are the squared population densities multiplied by the variance of the common measurement bias. The off-diagonal elements are the negative product of the two densities multiplied by the same variance. The resulting covariance matrix has rank one whenever the variance of the common measurement bias is positive and at least one density is nonzero.

The covariance induced by uncertainty in the common measurement bias is nonpositive: an increase in the transformed threshold decreases Se and increases Sp wherever the corresponding densities are positive. The variance of the common measurement bias and both marginal variance contributions remain nonnegative.

A.5. Calculation of the derivatives

The partial derivatives of each measure with respect to the six primitive inputs are obtained from its derivatives with respect to the three intermediate inputs by the chain rule.

$$\begin{aligned}
 a &= \text{Se}, & b &= \text{Sp}, \\
 J &= a + b - 1, \\
 q &= v a + (1 - v)(1 - b), \\
 p_o &= v a + (1 - v)b, \\
 p_e &= (1 - v) + (2v - 1)q, \\
 \eta &= v + q.
 \end{aligned} \tag{A.14}$$

$$\begin{aligned}
\text{PPV} &= \frac{v a}{q}, & \text{NPV} &= \frac{(1-v) b}{(1-q)}, \\
\text{FMI} &= a \sqrt{\frac{v}{q}}, \\
r_v &= \sqrt{v(1-v)}, & r_q &= \sqrt{q(1-q)}, \\
\text{MCC} &= \frac{r_v J}{r_q}.
\end{aligned} \tag{A.15}$$

The quantities q , p_o , and p_e denote predicted-positive probability, observed agreement, and chance agreement, respectively. The denominator in F_1 is denoted $\eta = v + q$; its square appears in the derivative denominators. For MCC, r_v and r_q denote the square-root factors associated with prevalence and predicted-positive probability, respectively; J retains its definition as Youden's index. PPV, NPV, and FMI retain their established definitions.

$$\begin{aligned}
q_a &= v, & q_b &= -(1-v), & q_v &= J; \\
p_{o_a} &= v, & p_{o_b} &= (1-v), & p_{o_v} &= a-b; \\
p_{e_a} &= v(2v-1), \\
p_{e_b} &= (1-v)(1-2v), \\
p_{e_v} &= -1 + 2q + (2v-1)J.
\end{aligned} \tag{A.16}$$

The derivatives of the three probabilities follow from their definitions. The derivative of predicted-positive probability with respect to Sp is nonpositive. The prevalence derivative of chance agreement includes both contributions from the product rule.

Table A4. All 48 elements of the 16×3 diagnostic-measure Jacobian

Measure	Jacobian element	Expanded scalar derivative
Se	$\frac{\partial \text{Se}}{\partial a}$	1
Se	$\frac{\partial \text{Se}}{\partial b}$	0
Se	$\frac{\partial \text{Se}}{\partial v}$	0
Sp	$\frac{\partial \text{Sp}}{\partial a}$	0
Sp	$\frac{\partial \text{Sp}}{\partial b}$	1
Sp	$\frac{\partial \text{Sp}}{\partial v}$	0
ACC	$\frac{\partial \text{ACC}}{\partial a}$	v
ACC	$\frac{\partial \text{ACC}}{\partial b}$	$(1-v)$
ACC	$\frac{\partial \text{ACC}}{\partial v}$	$a-b$
PPV	$\frac{\partial \text{PPV}}{\partial a}$	$\frac{v(1-v)(1-b)}{q^2}$
PPV	$\frac{\partial \text{PPV}}{\partial b}$	$\frac{v a(1-v)}{q^2}$

Measure	Jacobian element	Expanded scalar derivative
PPV	$\frac{\partial \text{PPV}}{\partial v}$	$\frac{a(1-b)}{q^2}$
NPV	$\frac{\partial \text{NPV}}{\partial a}$	$\frac{v(1-v)b}{(1-q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial b}$	$\frac{v(1-v)(1-a)}{(1-q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial v}$	$-\frac{b(1-a)}{(1-q)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial a}$	$\frac{b}{(1-b)(1-a)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial b}$	$\frac{a}{(1-a)(1-b)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial v}$	0
LR ⁺	$\frac{\partial \text{LR}^+}{\partial a}$	$\frac{1}{(1-b)}$
LR ⁺	$\frac{\partial \text{LR}^+}{\partial b}$	$\frac{a}{(1-b)^2}$
LR ⁺	$\frac{\partial \text{LR}^+}{\partial v}$	0
LR ⁻	$\frac{\partial \text{LR}^-}{\partial a}$	$-\frac{1}{b}$
LR ⁻	$\frac{\partial \text{LR}^-}{\partial b}$	$-\frac{(1-a)}{b^2}$
LR ⁻	$\frac{\partial \text{LR}^-}{\partial v}$	0
<i>J</i>	$\frac{\partial J}{\partial a}$	1
<i>J</i>	$\frac{\partial J}{\partial b}$	1
<i>J</i>	$\frac{\partial J}{\partial v}$	0
BA	$\frac{\partial \text{BA}}{\partial a}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial b}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial v}$	0
CZ	$\frac{\partial \text{CZ}}{\partial a}$	<i>b</i>
CZ	$\frac{\partial \text{CZ}}{\partial b}$	<i>a</i>
CZ	$\frac{\partial \text{CZ}}{\partial v}$	0

Measure	Jacobian element	Expanded scalar derivative
FMI	$\frac{\partial \text{FMI}}{\partial a}$	$\text{FMI} \left(\frac{1}{a} - \frac{v}{2q} \right)$
FMI	$\frac{\partial \text{FMI}}{\partial b}$	$\frac{\text{FMI} (1 - v)}{2q}$
FMI	$\frac{\partial \text{FMI}}{\partial v}$	$\text{FMI} \left(\frac{1}{2v} - \frac{J}{2q} \right)$
κ	$\frac{\partial \kappa}{\partial a}$	$\frac{(1 - p_e) v + (p_o - 1) v (2v - 1)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial b}$	$\frac{(1 - p_e) (1 - v) + (p_o - 1) (1 - v) (1 - 2v)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial v}$	$\frac{(1 - p_e) (a - b) + (p_o - 1) (-1 + 2q + (2v - 1) J)}{(1 - p_e)^2}$
PABAK	$\frac{\partial \text{PABAK}}{\partial a}$	$2v$
PABAK	$\frac{\partial \text{PABAK}}{\partial b}$	$2(1 - v)$
PABAK	$\frac{\partial \text{PABAK}}{\partial v}$	$2(a - b)$
F_1	$\frac{\partial F_1}{\partial a}$	$\frac{2v(v + (1 - v)(1 - b))}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial b}$	$\frac{2va(1 - v)}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial v}$	$\frac{2a(1 - b)}{\eta^2}$
MCC	$\frac{\partial \text{MCC}}{\partial a}$	$\frac{r_v}{r_q} \left(1 - \frac{J(1 - 2q)v}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial b}$	$\frac{r_v}{r_q} \left(1 + \frac{J(1 - 2q)(1 - v)}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial v}$	$\text{MCC} \left(\frac{(1 - 2v)}{2v(1 - v)} - \frac{J(1 - 2q)}{2q(1 - q)} \right)$

Rows follow the output order in equation (A.2). The three derivatives for each measure are taken with respect to a , b , and v . Denominators must be nonzero; expressions containing reciprocals or square roots require parameter values in their differentiability domains.

$$\mathbf{G} = \frac{\partial \mathbf{m}}{\partial \mathbf{z}^T} \in \mathbb{R}^{16 \times 3}, \quad (\text{A.17})$$

$$\mathbf{G}_{i \cdot} = \left(\frac{\partial F_i}{\partial a} \quad \frac{\partial F_i}{\partial b} \quad \frac{\partial F_i}{\partial v} \right) = (G_{ia} \quad G_{ib} \quad G_{iv}).$$

Table A4 supplies all elements of $\mathbf{G}$. The 16×1 output vector is differentiated with respect to the transpose of the 3×1 intermediate vector $\mathbf{z}$, yielding a 16×3 Jacobian.

A.5.1. Chain-rule expansion from three intermediate to six primitive inputs

$$\begin{aligned} a_m &= \frac{\partial a}{\partial m_D}, & a_s &= \frac{\partial a}{\partial s_D}, \\ b_m &= \frac{\partial b}{\partial m_H}, & b_s &= \frac{\partial b}{\partial s_H}. \end{aligned} \tag{A.18}$$

$$\mathbf{K} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}^T} = \begin{pmatrix} a_m & a_s & 0 & 0 & 0 & -f_D(t) \\ 0 & 0 & b_m & b_s & 0 & f_H(t) \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \tag{A.19}$$

The first row contains diseased-summary and transformed-threshold derivatives; the second contains nondiseased-summary and transformed-threshold derivatives. The third row selects prevalence. Population-family tail formulas supply the mean and standard-deviation derivatives. The transformed-threshold derivatives are minus the diseased density and plus the nondiseased density.

$$\begin{aligned} \mathbf{M} &= \mathbf{G} \mathbf{K} \in \mathbb{R}^{16 \times 6}, \\ \mathbf{M}_{i.} &= (G_{ia} a_m \quad G_{ia} a_s \quad G_{ib} b_m \quad G_{ib} b_s \quad G_{iv} \quad -G_{ia} f_D(t) + G_{ib} f_H(t)). \end{aligned} \tag{A.20}$$

Multiplication of $\mathbf{G}$ by $\mathbf{K}$ applies the chain rule to each measure. The sixth column of $\mathbf{M}$ combines the contributions of Se and Sp to the derivative with respect to the transformed threshold.

Population-summary derivatives are obtained by differentiating the fitted tail probabilities with respect to the arithmetic mean or standard deviation, including the induced changes in the distribution parameters. For the lognormal family, a change in the arithmetic standard deviation changes both lognormal parameters. Transformed-threshold derivatives are obtained directly from the population densities evaluated at the threshold.

A numerical approximation to a smooth scalar derivative is obtained by a central difference, with all other inputs held fixed:

$$\frac{g(x + h \mathbf{e}_j) - g(x - h \mathbf{e}_j)}{2h} \approx \partial g / \partial x_j.$$

The unit vector $\mathbf{e}_j$ identifies the perturbed input, and the positive step h has the same unit as that input. Both perturbed inputs must remain in the admissible domain. Stability under a reduction in step size provides a numerical check, although excessively small steps may amplify rounding effects.

A.6. Uncertainty propagation for a single measure

First-order propagation combines input variances and covariances with the corresponding partial derivatives to obtain the output variance. The combined standard uncertainty is its nonnegative square root. Variance contributions from independent sources are additive; standard uncertainties are not. The scalar expansion below expresses the calculation explicitly.

$$\mathbf{d} = \nabla g = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{pmatrix}, \quad (A.21)$$

$$\begin{aligned} d_1 &= \frac{\partial F}{\partial m_D}, & d_2 &= \frac{\partial F}{\partial s_D}, \\ d_3 &= \frac{\partial F}{\partial m_H}, & d_4 &= \frac{\partial F}{\partial s_H}, \\ d_5 &= \frac{\partial F}{\partial v}, & d_6 &= \frac{\partial F}{\partial t}. \end{aligned}$$

The gradient contains six partial derivatives. The propagated variance is the quadratic form obtained by multiplying the input covariance matrix by the gradient and premultiplying the result by its transpose.

$$\mathbf{w} = \Sigma_{\mathbf{x}} \mathbf{d} = \begin{pmatrix} A_D d_1 + B_D d_2 \\ B_D d_1 + C_D d_2 \\ A_H d_3 + B_H d_4 \\ B_H d_3 + C_H d_4 \\ u_v^2 d_5 \\ u_{\text{meas}}^2 d_6 \end{pmatrix}. \quad (A.22)$$

Each element of $\mathbf{w}$ is the scalar product of the corresponding row of the input covariance matrix and the gradient. The first element is $A_D d_1 + B_D d_2$, because the remaining elements of that covariance-matrix row are zero. Premultiplication by $\mathbf{d}^T$ gives the output variance.

$$u_c(F)^2 = \mathbf{d}^T \mathbf{w}, \quad (A.23)$$

$$u_c(F)^2 = A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2 + A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2 + u_v^2 d_5^2 + u_{\text{meas}}^2 d_6^2.$$

Symmetry of the population covariance matrices gives two equal covariance terms for each population. Their sums are $2B_D d_1 d_2$ and $2B_H d_3 d_4$ for the diseased and nondiseased populations, respectively. Cross-terms between independent sources are zero.

$$\begin{aligned} V_D &= A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2, \\ V_H &= A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2, \\ V_v &= u_v^2 d_5^2 = d_5^2 \frac{v(1-v)}{n_D + n_H}, \\ V_{\text{meas}} &= u_{\text{meas}}^2 d_6^2 = d_6^2 [b_0^2 + (b_1 t)^2], \\ u_c(F)^2 &= V_D + V_H + V_v + V_{\text{meas}}. \end{aligned} \quad (A.24)$$

Equation (A.24) decomposes the output variance into source-specific contributions and retains the covariance between the estimated mean and standard deviation within each population. Sampling variance includes the population-parameter contributions and, when included, the prevalence contribution. The measurement variance contribution results from propagating uncertainty in the common measurement bias while holding the diagnostic threshold fixed. The combined standard uncertainty is the nonnegative square root of their sum. The measurement-variance expression in equation (A.24) uses the root-sum-of-squares model.

Within-population covariance terms may be positive or negative and are retained in the corresponding variance contributions. Each valid population covariance matrix yields a nonnegative propagated variance, although a covariance term may reduce that variance. Omitting a negative covariance term changes the assumed dependence structure.

For a generic measure F , combined, total sampling, and measurement standard uncertainties are denoted $u_c(F)$, $u_s(F)$, and $u_m(F)$, respectively; the parameter and prevalence sampling components are $u_{\text{par}}(F)$ and $u_{\text{prev}}(F)$. Figure legends use this notation, with the measure argument implicit where omitted.

The combined variance is the sum of the independent source-specific variance contributions, with all within-population covariances retained. Parameter sampling uncertainty comprises the diseased and nondiseased population contributions; total sampling uncertainty also includes the prevalence contribution.

$$u_s(F)^2 = u_{\text{par}}(F)^2 + u_{\text{prev}}(F)^2, \quad u_c(F)^2 = u_s(F)^2 + u_m(F)^2$$

A.7. Joint uncertainty propagation for all sixteen measures

$$\begin{aligned} V_a &= A_D a_m^2 + 2B_D a_m a_s + C_D a_s^2 + f_D(t)^2 u_{\text{meas}}^2, \\ V_b &= A_H b_m^2 + 2B_H b_m b_s + C_H b_s^2 + f_H(t)^2 u_{\text{meas}}^2, \\ C_{ab} &= -f_D(t) f_H(t) u_{\text{meas}}^2. \end{aligned} \tag{A.25}$$

The first three terms of V_a give the contribution from the estimated diseased population mean and standard deviation to the variance of Se; the final term gives the measurement contribution. V_b contains the corresponding nondiseased population and measurement contributions to the variance of Sp. Under the stated independence assumptions, C_{ab} arises solely from uncertainty in the common measurement bias and is nonpositive because the two transformed-threshold derivatives have opposite signs.

$$\begin{aligned} \Sigma_z &= \mathbf{K} \Sigma_x \mathbf{K}^T, \\ \Sigma_z &= \begin{pmatrix} V_a & C_{ab} & 0 \\ C_{ab} & V_b & 0 \\ 0 & 0 & u_v^2 \end{pmatrix}. \end{aligned} \tag{A.26}$$

The intermediate covariance matrix has dimensions 3×3 . Covariances involving prevalence are zero under its assumed independence from the fitted population summaries and the common measurement bias. Other sampling designs may require additional covariance terms.

$$\begin{aligned} \Sigma_m &= \mathbf{M} \Sigma_x \mathbf{M}^T, \\ \Sigma_m &= \mathbf{G} \Sigma_z \mathbf{G}^T. \end{aligned} \tag{A.27}$$

The two expressions yield the same 16×16 output covariance matrix because $\mathbf{M} = \mathbf{G} \mathbf{K}$. They represent equivalent formulations of uncertainty propagation. The factorization distinguishes the 16×3 Jacobian with respect to intermediate inputs from the 16×6 Jacobian with respect to primitive inputs.

$$\begin{aligned} \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ia} 0 G_{jv}, \\ &\quad + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{ib} 0 G_{jv}, \\ &\quad + G_{iv} 0 G_{ja} + G_{iv} 0 G_{jb} + G_{iv} u_v^2 G_{jv}, \\ \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{iv} u_v^2 G_{jv}. \end{aligned} \tag{A.28}$$

The expanded product contains nine terms, four of which vanish under the assumed independence of prevalence. Off-diagonal elements are covariances between measures, and diagonal elements are their variances.

$$u_c(F_i)^2 = G_{ia}^2 V_a + 2G_{ia} G_{ib} C_{ab} + G_{ib}^2 V_b + G_{iv}^2 u_v^2. \tag{A.29}$$

The factor of two in equation (A.29) arises from the two equal covariance terms for Se and Sp. When C_{ab} is negative, their combined contribution decreases the propagated variance when the corresponding partial derivatives have the same sign and increases it when they have opposite signs. Omitting this contribution would represent the common measurement contributions to Se and Sp as independent and would contradict the stated measurement model.

$$\begin{aligned} V_{\text{meas}_i} &= u_{\text{meas}}^2 \left(-G_{ia} f_D(t) + G_{ib} f_H(t) \right)^2, \\ V_{\text{meas}_i} &= u_{\text{meas}}^2 \left[G_{ia}^2 f_D(t)^2 - 2 G_{ia} G_{ib} f_D(t) f_H(t) + G_{ib}^2 f_H(t)^2 \right]. \end{aligned} \quad (\text{A.30})$$

Expansion of the squared derivative with respect to the transformed threshold recovers the covariance contribution for Se and Sp already included in the propagated variance.

A.7.1. Source-specific expansion in primitive coordinates

$$\Sigma_{\mathbf{m}} = \mathbf{M}_D \Sigma_D \mathbf{M}_D^T + \mathbf{M}_H \Sigma_H \mathbf{M}_H^T + \mathbf{j}_v u_v^2 \mathbf{j}_v^T + \mathbf{j}_{\text{meas}} u_{\text{meas}}^2 \mathbf{j}_{\text{meas}}^T. \quad (\text{A.31})$$

The diseased and nondiseased Jacobian submatrices comprise the first two and the next two columns of $\mathbf{M}$, respectively. The prevalence and transformed-threshold vectors $\mathbf{j}_v$ and $\mathbf{j}_{\text{meas}}$ are its fifth and sixth columns. Each source-specific covariance contribution has dimensions 16×16 .

$$\begin{aligned} \Sigma_{\mathbf{m}_{ij}} &= A_D M_{i1} M_{j1} + B_D (M_{i1} M_{j2} + M_{i2} M_{j1}) + C_D M_{i2} M_{j2}, \\ &+ A_H M_{i3} M_{j3} + B_H (M_{i3} M_{j4} + M_{i4} M_{j3}) + C_H M_{i4} M_{j4}, \\ &+ u_v^2 M_{i5} M_{j5} + u_{\text{meas}}^2 M_{i6} M_{j6}. \end{aligned} \quad (\text{A.32})$$

The expanded expression retains both within-population covariance terms. Equal output indices recover the scalar quadratic form, including the factors of two.

The diagonal elements of the output covariance matrix are the variances of the sixteen measures; the off-diagonal elements are their pairwise covariances. The scalar and joint formulations yield identical variances. Their agreement provides a check of implementation consistency but does not assess the fit of the population model to the observations.

A.8. Conversion of standard uncertainty to an approximate expanded-uncertainty interval

Standard uncertainty quantifies the scale of uncertainty, whereas expanded uncertainty defines an interval half-width. The coverage factor depends on the specified nominal coverage level and the approximate effective degrees of freedom. Each population's variance contribution retains the covariance between its estimated mean and standard deviation in the calculation of effective degrees of freedom.

The four nonnegative source-specific variance contributions are denoted V_D , V_H , V_v , and V_{meas} , with the covariance between each population's estimated mean and standard deviation retained. Effective degrees of freedom are approximated by [20, 21]:

$$\begin{aligned} V_D &= \nabla g_D^T \Sigma_D \nabla g_D, \\ V_H &= \nabla g_H^T \Sigma_H \nabla g_H, \\ V_v &= g_v^2 u_v^2, \\ V_{\text{meas}} &= g_t^2 u_{\text{meas}}^2. \\ v_{\text{eff}} &= \frac{(V_D + V_H + V_v + V_{\text{meas}})^2}{\frac{V_D^2}{v_D} + \frac{V_H^2}{v_H} + \frac{V_v^2}{v_v} + \frac{V_{\text{meas}}^2}{v_{\text{meas}}}}. \end{aligned}$$

The sampling degrees of freedom are $v_D = n_D - 1$, $v_H = n_H - 1$, and $v_v = n_D + n_H - 1$. Assigning $v_{\text{meas}} = \infty$ treats the standard uncertainty of the common measurement bias as known without appreciable uncertainty. This assumption does not follow solely from classification as a Type B evaluation. For a nominal coverage level of $1 - \alpha$:

$$k = t_{v_{\text{eff}}, 1-\alpha/2}, \quad U = k u_c(F)$$

Here k is the $1 - \alpha/2$ quantile of Student's t distribution with v_{eff} degrees of freedom, and U is the expanded uncertainty. If the measurement uncertainty component has infinite degrees of freedom, its term in the

denominator of the effective-degrees-of-freedom expression is zero, but its contribution to the combined variance remains.

The two-sided expanded-uncertainty interval is:

$$I_{1-\alpha}(F) = [F - U, F + U]$$

At specified thresholds, these pointwise expanded-uncertainty intervals are based on a first-order Wald approximation. At optimized thresholds, common measurement uncertainty is propagated nonlinearly by Gauss–Hermite quadrature, conditional on the estimated optimal threshold. The notebook permits restriction of interval endpoints to each measure's admissible range and retains the unrestricted intervals for verification. Restriction to the admissible range does not ensure nominal coverage or remove the limitations of the approximation. Standard uncertainties are unaffected by this restriction.

A.9. Evaluation of uncertainty at an estimated optimal threshold

In *DiagAccU.nb*, Wolfram NMaximize maximizes the selected measure's point estimate subject to $t > m_H$, without a prescribed upper threshold bound. The fitted population distributions remain fixed. At the estimated optimal threshold, first-order sampling variance is combined with measurement variance obtained by nine-point Gauss–Hermite quadrature under a centered normal model for the common measurement bias.

The resulting interval quantifies uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold subject to $t > m_H$. It excludes uncertainty associated with threshold estimation and selection.

At a smooth interior maximum, the first derivative with respect to the threshold is zero. First-order propagation may therefore yield a zero contribution from the common measurement bias, although its nonlinear effect changes the measure. Nonlinear propagation accounts for these changes through weighted evaluations of the complete measure at quadrature nodes.

The estimated optimal cutoff is held fixed. A centered normal model for the common measurement bias induces a normal transformed threshold with mean equal to that cutoff and standard deviation equal to the standard uncertainty of the shift. Nine-point Gauss–Hermite quadrature approximates expectations by nine weighted function evaluations. For nodes and weights defined with respect to the weight $\exp(-x^2)$, the transformed-threshold nodes and normalized weights are:

$$\tau_i = t^* + \sqrt{2} u_{\text{meas}}(t^*) x_i, \quad p_i = w_i / \sqrt{\pi}, \quad i = 1, \dots, 9.$$

The complete diagnostic accuracy measure is evaluated at each quadrature node, with the same transformed threshold applied to both population distributions and the fitted population summaries and prevalence held fixed. The standard uncertainty of the common measurement bias is evaluated at the estimated optimal threshold and held fixed throughout the integration. The weighted variance of the nine evaluations gives the common measurement variance contribution. Its sum with the first-order sampling variance gives the combined variance, whose nonnegative square root is the combined standard uncertainty. Centered normal symmetry makes either sign of the quadrature displacement equivalent.

The estimated optimal threshold remains the center of the conditional analysis. The weighted mean used to calculate the common measurement variance may differ from the measure evaluated at that threshold. Population distributions are not re-estimated, and threshold selection is not repeated. Assessment of threshold-selection uncertainty requires repetition of these estimation and selection steps, for example in an appropriately specified bootstrap. For positive-support families, transformed-threshold evaluations follow the support conventions of the fitted distributions.

A.10. Implementation and reproducibility

A single AroundReplace operation propagates the final scalar measure from a VectorAround object for the four correlated population summaries, an optional Around input for prevalence, and one Around input for the transformed threshold [25, 26]. DiagnosticAccuracyAllComponents, DiagnosticAccuracyAllConfidenceIntervals,

and `DiagnosticAccuracyJointResult` provide reproducible component, interval, and covariance records, respectively. Existing software argument and result labels are retained for compatibility.

DiagAccU.nb is the computational source. *DiagAccUCalculations.nb* presents the symbolic calculations, and the *Interface Documentation* describes software controls and options. Sections A.4–A.7 integrate the expanded covariance products and all 48 intermediate derivatives [43–45].

Implementation checks comprise agreement between each scalar variance and the corresponding diagonal element of the joint output covariance matrix, agreement between the sum of the four source-specific covariance contributions and the total, symmetry, and nonnegative source-specific variances. These algebraic checks assess internal consistency. Interval coverage, model fit, transportability, and uncertainty associated with threshold selection require separate assessment.

Appendix B. Abbreviations, notation, and software

B.1. List of abbreviations

DAM: diagnostic accuracy measure

CI: confidence interval; the software label for an approximate pointwise expanded-uncertainty interval

GUM: Guide to the Expression of Uncertainty in Measurement

OGTT: oral glucose tolerance test

FPG: fasting plasma glucose

ROC: receiver operating characteristic

B.2. Notation

B.2.1. Populations

H: nondiseased population

D: diseased population

B.2.2. Test outcomes

TN: true negative

TP: true positive

FN: false negative

FP: false positive

B.2.3. DAMs

Se: sensitivity

Sp: specificity

PPV: positive predictive value

NPV: negative predictive value

ACC: overall diagnostic accuracy

DOR: diagnostic odds ratio

LR⁺: positive likelihood ratio

LR⁻: negative likelihood ratio

J: Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa

F_1 : F_1 score

MCC: Matthews correlation coefficient

B.2.4. Parameters

μ_P, m_D, m_H : population arithmetic mean and supplied sample means, respectively; P denotes D or H

σ_P, s_D, s_H : population standard deviation and supplied Bessel-corrected sample standard deviations, respectively

n_D, n_H : sample sizes of the diseased and nondiseased populations, respectively

v : prevalence

t : specified fixed diagnostic cutoff

Δ : common measurement bias; $\tau = t - \Delta$: transformed threshold, with nominal value t ; $u_{\text{meas}}(t)$: standard uncertainty of the common shift at the fixed cutoff

t^* : optimal threshold subject to $t > m_H$

$1 - \alpha$: nominal confidence level

B.2.5. Functions and relations

$u_s(F)$: total sampling standard uncertainty of measure F

$u_m(F)$: common measurement standard uncertainty of F

$u_c(F)$: combined standard uncertainty of F

$V_D, V_H, V_v, V_{\text{meas}}$: variance contributions from independent uncertainty sources

$f_D(t), f_H(t)$: population-specific probability densities evaluated at the diagnostic threshold

$F_D(t), F_H(t)$: population-specific cumulative distribution functions evaluated at the diagnostic threshold

q : predicted-positive probability; p_o : observed agreement; p_e : chance agreement

Σ_x : covariance matrix of the six primitive inputs; J : output Jacobian

$I_{1-\alpha}(F)$: approximate expanded-uncertainty interval for F

$\text{Var}(F)$: variance of F ; $\text{Cov}(A, B)$: covariance of A and B

ν_{eff} : effective degrees of freedom calculated from the variance contributions of independent uncertainty sources

k : coverage factor

$U = k u_c$: expanded uncertainty

B.3. Input

B.3.1. Range of input parameters

Most threshold controls use $t_{\min} = \max(0.01, \min(m_H - 6s_H, m_D - 6s_D))$ and $t_{\max} = \max(m_H + 6s_H, m_D + 6s_D)$. The selected-measure standard-uncertainty table across all distribution pairs (tdpu) instead uses $\max(m_H - 5s_H, 0) \leq t \leq m_D + 5s_H$.

n_D : 2 – 10,000

m_D : 0.1 – 10,000

s_D : 0.01 – 1,000

$n_{\bar{H}}$: 2 – 10,000

$m_{\bar{H}}$: 0.1 – 10,000

$s_{\bar{H}}$: 0.01 – 1,000

v : 0.001 – 0.999

n_U : 20–10,000.

b_0 : $0 \leq b_0 \leq s_H$

b_1 : 0–0.1

$1 - \alpha$: 0.900–0.999

The threshold t , both population means m_D and m_H , both standard deviations s_D and s_H , and b_0 must use the same measurement unit; b_1 is dimensionless.

n : 20–20,000 in views with a total-sample-size control. In the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

B.3.2. Additional input options

B.3.2.1. Plots

Extended and limited settings change the profile domain. Their limits depend on the selected view; they do not define the optimization domain or establish interval coverage.

B.3.2.2. Tables

Tables provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

B.3.3. Program controls

The program uses a tabbed interface for entering inputs and viewing estimates, uncertainty components, and plots.

Users may define the numerical settings with menus or sliders. Sliders are finely manipulated by pressing the *alt* or *opt* key while dragging the mouse. Pressing the *shift* or *ctrl* keys can even more finely manipulate them.

Plot interaction depends on the Wolfram Front End and the selected view. The interface documentation describes the available controls and profile domains.

B.4. Software availability and requirements

- a) Program name: *DiagAccU*
- b) Version: 2.0.1
- c) DOI: [10.5281/zenodo.22843673](https://doi.org/10.5281/zenodo.22843673)
- d) Project home page: <https://www.hcsl.com/Tools/DiagnosticAccuracy/>
- e) Computational source notebook: *DiagAccU.nb*. Available at:
- f) <https://www.hcsl.com/Tools/DiagnosticAccuracy/DiagAccU.nb>
- g) Programming language: Wolfram Language
- h) Operating systems: Microsoft Windows 10+, Linux 4.18+, Apple macOS 13+
- i) Hardware requirements: x86-64-v2 and higher or Apple Silicon microarchitecture CPU, and 16 GB+ RAM, recommended.
- j) Software citation: Chatzimichail RA, Chatzimichail T, Hatjimihail AT. *DiagAccU*: a unified framework for uncertainty estimation of diagnostic accuracy measures. Version 2.0.1. Hellenic Complex Systems Laboratory; 2026. DOI: <https://doi.org/10.5281/zenodo.22843673>.
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